

LIMOGES AP, France

WMO: 074340

Lat: 45.8611N

Lon: 1.1750E

Elev: 1300

StdP: 14.02

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	25.5	13.9	11.7	30.2	18.1	14.4	30.8	23.2	43.5	21.2	43.3	7.8	30	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	87.1	68.1	83.4	67.1	80.2	65.5	70.2	83.0	68.5	80.4	67.0	77.8	7.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	100.6	75.5	64.4	94.9	73.6	62.9	89.9	71.8	34.9	83.0	33.5	80.4	32.1	77.9	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.6	17.3	15.5	19.1	92.3	3.9	3.4	16.3	94.7	14.0	96.7	11.8	98.6	8.9	101.1		
			18.0	73.2	4.1	2.2	15.0	74.8	12.6	76.0	10.3	77.3	7.3	78.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	40.0	40.8	46.5	50.7	56.8	63.3	66.6	66.8	60.7	55.1	45.6	41.3
	DBStd	11.75	6.59	7.34	6.76	6.72	6.80	7.10	6.47	6.61	6.09	6.66	6.55	6.92
	HDD50	1277	313	264	149	71	15	1	0	0	2	29	160	273
	HDD65	4740	776	678	573	432	267	116	59	55	158	311	581	734
	CDD50	2344	2	6	41	91	227	400	515	520	323	186	29	4
	CDD65	330	0	0	0	1	14	65	108	111	28	3	0	0
	CDH74	2672	0	0	0	14	104	540	885	930	190	9	0	0
	CDH80	763	0	0	0	0	9	144	257	320	33	0	0	0
Wind	WSAvg	7.7	8.4	8.6	8.7	8.4	7.5	7.1	6.9	6.3	6.9	7.3	7.9	8.2
Precipitation	PrecAvg	39.4	3.6	3.1	3.2	3.3	3.7	2.9	2.4	2.9	3.0	3.3	3.8	3.9
	PrecMax	57.0	7.7	8.7	8.7	7.4	7.8	6.3	6.5	8.2	8.7	6.3	8.9	9.8
	PrecMin	28.7	1.0	0.2	0.2	0.2	0.8	0.0	0.2	0.4	0.3	0.2	0.7	0.1
	PrecStd	5.6	1.8	1.8	1.9	1.9	1.5	1.5	1.4	1.7	1.9	1.7	1.8	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	63.0	69.6	76.0	81.1	89.8	91.4	93.6	84.0	75.0	64.5	57.4
		MCWB	48.2	49.3	53.1	58.2	64.3	69.8	68.6	69.1	64.6	62.4	55.6	50.5
	2%	DB	52.7	57.1	64.7	70.5	76.9	83.8	86.1	87.5	79.1	70.9	60.3	54.4
		MCWB	48.7	47.3	51.5	56.1	61.3	67.7	68.2	67.8	63.0	61.0	53.5	49.6
	5%	DB	50.7	53.1	60.8	66.5	73.1	80.2	82.4	83.2	75.4	67.9	57.3	52.1
		MCWB	47.6	46.5	50.1	54.0	60.2	65.8	67.0	67.1	62.3	59.7	52.8	48.8
	10%	DB	48.5	49.9	56.9	62.4	69.1	76.4	78.9	79.1	71.5	64.5	54.7	50.1
		MCWB	45.9	45.9	48.2	52.0	58.5	64.4	65.3	65.5	60.9	58.2	51.4	47.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	51.8	55.1	60.1	66.6	72.6	71.7	71.9	68.3	64.7	57.6	53.0
		MCDB	53.0	57.9	64.4	72.6	77.7	83.8	84.6	85.3	77.4	71.2	61.3	55.1
	2%	WB	50.1	49.5	52.9	57.5	63.8	69.4	69.8	70.0	65.8	62.5	55.4	51.2
		MCDB	51.7	53.7	61.4	68.5	74.0	80.9	82.8	83.7	75.2	68.2	58.7	53.1
	5%	WB	48.1	47.9	51.1	55.1	61.5	67.3	68.1	68.3	63.7	60.8	53.5	49.6
		MCDB	50.0	51.2	58.0	64.1	70.4	78.0	79.8	80.4	72.1	65.9	56.4	51.7
	10%	WB	46.4	46.4	49.4	52.9	59.2	65.1	66.4	66.5	61.9	59.1	51.7	47.9
		MCDB	48.1	49.3	55.1	60.6	67.3	74.7	77.2	77.1	69.4	63.9	54.5	49.7
Mean Daily Temperature Range	5% DB	MDBR	9.1	11.3	13.8	14.9	15.9	16.5	16.9	17.0	15.9	13.1	10.3	9.3
		MCDBR	11.0	16.5	19.5	20.9	21.2	21.5	21.7	22.3	21.1	16.6	13.5	11.4
		MCWBR	8.1	10.2	10.6	11.0	11.1	10.6	9.7	9.4	9.8	9.4	8.9	8.2
	5% WB	MCDBR	8.8	12.5	16.5	18.7	19.1	20.0	20.2	19.9	18.3	14.2	11.8	9.4
		MCWBR	7.4	8.9	9.8	10.3	11.1	10.5	10.1	9.3	9.8	8.6	8.4	7.6
Clear-Sky Solar Irradiance	taub	0.303	0.319	0.348	0.378	0.380	0.393	0.384	0.379	0.349	0.345	0.319	0.300	
	taud	2.494	2.445	2.375	2.289	2.329	2.326	2.346	2.373	2.436	2.459	2.487	2.492	
	Ebn at Noon	257	273	279	279	281	277	278	275	275	259	247	244	
	Edh at Noon	21	27	33	39	39	39	38	36	31	26	21	19	
All-Sky Solar Radiation	RadAvg	384	665	1033	1413	1654	1909	1866	1648	1304	813	456	345	
	RadStd	46	87	129	182	150	176	130	136	85	68	57	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page